Supplemental Material for "Fast and Efficient Inference for Flexible Max-Stable Processes"

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A Truncated extremal-t processes

A.1 Proof of Theorem 2

By applying the first half of the equation (8) in the main manuscript we have

$$
\begin{aligned}
V(\boldsymbol{x}) &= \sum_{k=1}^{D} \tfrac{1}{a_k x_k} \int_0^\infty u_k^\nu \int_{\mathbf{0}}^{u_k\left(\frac{\boldsymbol{x}^\circ_{-k}}{x_k^\circ}\right)^{1/\nu}} \tfrac{\phi(\boldsymbol{u};\mathbf{0},\Sigma)}{\Phi^\circ(\boldsymbol{\infty};\mathbf{0},\Sigma)} \mathrm{d}\boldsymbol{u}_{-k}\mathrm{d}u_k \\
&= \sum_{k=1}^{D} \tfrac{1}{a_k x_k \Phi^\circ(\boldsymbol{\infty};\mathbf{0},\Sigma)} \int_0^\infty u_k^\nu \phi(u_k;0,1) \int_{\mathbf{0}}^{u_k\left(\frac{\boldsymbol{x}^\circ_{-k}}{x_k^\circ}\right)^{1/\nu}} \phi(\boldsymbol{u}_{-k};\Sigma_{-k,k}u_k, \mathbb{I}_{D-1} - \Sigma_{-k,k}\Sigma_{k,-k})\mathrm{d}\boldsymbol{u}_{-k}\mathrm{d}u_k \\
&= \sum_{k=1}^{D} \tfrac{1}{a_k x_k \Phi^\circ(\boldsymbol{\infty};\mathbf{0},\Sigma)} \int_0^\infty u_k^\nu \phi(u_k;0,1)\Phi^\circ\left(u_k\left(\frac{\boldsymbol{x}^\circ_{-k}}{x_k^\circ}\right)^{1/\nu}; \Sigma_{-k,k}u_k, \mathbb{I}_{D-1} - \Sigma_{-k,k}\Sigma_{k,-k}\right)\mathrm{d}u_k \\
&= \sum_{k=1}^{D} \tfrac{2^{(\nu-2)/2}\Gamma((\nu+1)/2)}{\Phi^\circ(\boldsymbol{\infty};\mathbf{0},\Sigma) a_j x_j \sqrt{\pi}} T^\circ_{\nu+1}\left(\left(\frac{\boldsymbol{x}^\circ_{-k}}{x_k^\circ}\right)^{1/\nu}; \Sigma_{-k,k}, \tfrac{\mathbb{I}_{D-1}-\Sigma_{-k,k}\Sigma_{k,-k}}{\nu+1}\right),
\end{aligned} \tag{1}
$$

where $\boldsymbol{x}^\circ = \boldsymbol{x}\boldsymbol{a}$, with $\boldsymbol{a} = (a_1, \ldots, a_D)$, $\mathbb{I}_{D-1}$ is the $(D-1)$-dimensional identity matrix, and the index $-k$ represents the set $B_D \setminus \{k\}$.

The normalising constants a_k can be either derived from the fact that $V(\infty, \ldots, \infty, x_k, \infty, \ldots, \infty) = 1/x_k$ or by comparing the integral above to the following integral

$$
a_k = E[\tilde{Y}_k^\nu] = \int_0^\infty u_k^\nu \int_{\mathbf{0}}^{\boldsymbol{\infty}} \tfrac{\phi(\boldsymbol{u};\mathbf{0},\Sigma)}{\Phi^\circ(\boldsymbol{\infty};\mathbf{0},\Sigma)} \mathrm{d}\boldsymbol{u}_{-k}\mathrm{d}u_k.
$$

The use of either way will lead us to

$$a_k = \frac{2^{(\nu-2)/2}\Gamma((\nu+1)/2)}{\Phi^\circ(\boldsymbol{\infty};\mathbf{0},\Sigma)\sqrt{\pi}} T^\circ_{\nu+1}\left(\boldsymbol{\infty}_{D-1}; \Sigma_{-k,k}, \frac{\mathbb{I}_{D-1}-\Sigma_{-k,k}\Sigma_{k,-k}}{\nu+1}\right).$$

Plugging the expression for a_k in (1), we have

$$V(\boldsymbol{x}) = \sum_{k=1}^{D} \frac{T^\circ_{\nu+1}\left(\left(\frac{\boldsymbol{x}^\circ_{-k}}{x^\circ_k}\right)^{1/\nu}; \Sigma_{-k,k}, \frac{\mathbb{I}_{D-1}-\Sigma_{-k,k}\Sigma_{k,-k}}{\nu+1}\right)}{x_k T^\circ_{\nu+1}\left(\boldsymbol{\infty}_{D-1}; \Sigma_{-k,k}, \frac{\mathbb{I}_{D-1}-\Sigma_{-k,k}\Sigma_{k,-k}}{\nu+1}\right)}.$$

A.2 Proof of Proposition 1

From the distribution of $\tilde{\boldsymbol{Y}}$ and therefore $\boldsymbol{W}$, the conditional probability in (12) from the manuscript is written as

$$\begin{aligned}\Pr(\boldsymbol{W}_{\overline{B}_k} = \mathbf{0}_{D-k} \mid \boldsymbol{W}_{B_k} = r\boldsymbol{x}_{B_k}) &= \Phi^\circ\left(\mathbf{0}_{D-k}; \Sigma_{\overline{B}_k}\Sigma^{-1}_{B_kB_k}\left(r\boldsymbol{x}^\circ_{B_k}\right)^{1/\nu}, \Sigma_{\overline{B}_k\overline{B}_k} - \Sigma_{\overline{B}_kB_k}\Sigma^{-1}_{B_kB_k}\Sigma_{\overline{B}_kB_k}\right)\\ &= 0,\end{aligned}$$

which, by Theorem 1, implies that the truncated extramal-t process has no mass on $\partial\Omega$. It now remains to derive its density on Ω°.

Take $\boldsymbol{w} = (w_1, \ldots, w_D) \in \mathbb{R}^D_+$ and let $\boldsymbol{w}^\circ = \boldsymbol{w}\boldsymbol{a}$, the density of $\boldsymbol{W}$ is given by

$$\begin{aligned} f_W(\boldsymbol{w}) &= f_{\tilde{\boldsymbol{Y}}}\left((\boldsymbol{w}^\circ)^{1/\nu}\right) \times \prod_{k=1}^{D}\left\{\frac{a_k}{\nu}\left(w^\circ_k\right)^{1/\nu-1}\right\}\\ &= \frac{\phi\left((\boldsymbol{w}^\circ)^{1/\nu};\mathbf{0},\Sigma\right)}{\Phi^\circ(\boldsymbol{\infty};\mathbf{0},\Sigma)} \times \prod_{k=1}^{D}\left\{\frac{a_k}{\nu}\left(w^\circ_k\right)^{1/\nu-1}\right\}. \end{aligned} \tag{2}$$

In order to complete the proof we need the following lemma.

Lemma 1. *The intensity function for the truncated extremal-t process is*

$$\kappa(\boldsymbol{x}) = \frac{\prod_{k=1}^{D}\left(a_k x_k^{1-\nu}\right)^{1/\nu}}{\Phi^\circ(\boldsymbol{\infty};\mathbf{0},\Sigma)\nu^{D-1}|\Sigma|^{1/2}\pi^{D/2}2^{(2-\nu)/2}}\left\{\left[(\boldsymbol{x}^\circ)^{1/\nu}\right]^\top \Sigma^{-1}\left[(\boldsymbol{x}^\circ)^{1/\nu}\right]\right\}^{-(\nu+D)/2}\Gamma\left(\tfrac{\nu+D}{2}\right).$$

Proof. By plugging (2) into (9) from the main manuscript, we get

$$\begin{aligned}
\kappa(\boldsymbol{x}) &= \int_0^\infty r^D \frac{\phi((r\boldsymbol{x}^\circ)^{1/\nu};\mathbf{0},\Sigma)}{\Phi^\circ(\infty;\mathbf{0},\Sigma)} \prod_{k=1}^D \left\{\frac{a_k}{\nu}(rx_k^\circ)^{1/\nu-1}\right\} \mathrm{d}r \\
&= \frac{\prod_{k=1}^D \left\{a_k^{1/\nu} x_k^{1/\nu-1}\right\}}{\Phi^\circ(\infty;\mathbf{0},\Sigma)\nu^D} \int_0^\infty r^{D/\nu}\phi((r\boldsymbol{x}^\circ)^{1/\nu};\mathbf{0},\Sigma)\mathrm{d}r \\
&= \frac{\prod_{k=1}^D \left(a_k x_k^{\nu-1}\right)^{1/\nu}}{\Phi^\circ(\infty;\mathbf{0},\Sigma)\nu^D|\Sigma|^{1/2}\pi^{D/2}2^{(2-\nu)/2}} \int_0^\infty r^{D/\nu} \exp\left\{-\frac{r^{2/\nu}}{2}\left[(\boldsymbol{x}^\circ)^{1/\nu}\right]^\top \Sigma^{-1}\left[(\boldsymbol{x}^\circ)^{1/\nu}\right]\right\} \mathrm{d}r \\
&= \frac{\prod_{k=1}^D \left(a_k x_k^{1-\nu}\right)^{1/\nu}}{\Phi^\circ(\infty;\mathbf{0},\Sigma)\nu^{D-1}|\Sigma|^{1/2}\pi^{D/2}2^{(2-\nu)/2}} \left\{\left[(\boldsymbol{x}^\circ)^{1/\nu}\right]^\top \Sigma^{-1}\left[(\boldsymbol{x}^\circ)^{1/\nu}\right]\right\}^{-(\nu+D)/2} \Gamma\left(\tfrac{\nu+D}{2}\right),
\end{aligned}$$

which completes the proof. □

By Lemma 1, the density on the interior of the simplex, Ω° is therefore given by

$$h(\boldsymbol{w}) = \frac{\prod_{k=1}^D \left(a_k w_k^{1-\nu}\right)^{1/\nu}}{\Phi^\circ(\infty;\mathbf{0},\Sigma)\nu^{D-1}|\Sigma|^{1/2}\pi^{D/2}2^{(2-\nu)/2}} \left\{\left[(\boldsymbol{w}^\circ)^{1/\nu}\right]^\top \Sigma^{-1}\left[(\boldsymbol{w}^\circ)^{1/\nu}\right]\right\}^{-(\nu+D)/2} \Gamma\left(\tfrac{\nu+D}{2}\right), \quad \boldsymbol{w} = \tfrac{\boldsymbol{x}}{\|\boldsymbol{x}\|}.$$

A.3 Partial derivatives of the exponent function for the truncated extremal-t processes

Lemma 2. *The partial derivatives of the exponent function of the truncated extremal-t process are given by*

$$\begin{aligned}
-V_{B_k}(\boldsymbol{x}) = &\frac{\prod_{i\in B_k}\left\{a_i x_i^{1-\nu}\right\}^{1/\nu} 2^{(\nu-2)/2}\Gamma\left(\frac{k+\nu}{2}\right)\nu^{1-k}}{\Phi^\circ(\infty;\mathbf{0},\Sigma)\pi^{k/2}|\Sigma_{B_kB_k}|^{1/2} Q_{\Sigma_{B_kB_k}}\left(\boldsymbol{x}_{B_k}^\circ\right)^{k+\nu}} \\
&\times T_{\nu+k}^\circ\left(\frac{\left(\boldsymbol{x}_{B_k}^\circ\right)^{1/\nu}}{Q_{\Sigma_{B_kB_k}}\left(\boldsymbol{x}_{B_k}^\circ\right)}; \frac{\Sigma_{\overline{B}_kB_k}\Sigma_{B_kB_k}^{-1}\left(\boldsymbol{x}_{B_k}^\circ\right)^{1/\nu}}{Q_{\Sigma_{B_kB_k}}\left(\boldsymbol{x}_{B_k}^\circ\right)}, \frac{\Sigma_{\overline{B}_k\overline{B}_k}-\Sigma_{\overline{B}_kB_k}\Sigma_{B_kB_k}^{-1}\Sigma_{\overline{B}_kB_k}}{k+\nu}\right),
\end{aligned}$$

where $Q_\Sigma(\boldsymbol{x}) = \left\{\boldsymbol{x}^\top\Sigma^{-1}\boldsymbol{x}\right\}^{1/2}$.

Proof. Since the interest is in the partial derivatives of the exponent function $V(\boldsymbol{x})$ with respect to the first k components, i.e., $\boldsymbol{x}_{B_k}$, the correlation matrix σ is decomposed as

$$\Sigma = \begin{bmatrix} \Sigma_{B_kB_k} & \Sigma_{B_k\overline{B}_k} \\ \Sigma_{\overline{B}_kB_k} & \Sigma_{\overline{B}_k\overline{B}_k} \end{bmatrix}.$$

Plugging (2) into (11) from the main manuscript, we have

$$
\begin{aligned}
-V_{B_k}(\boldsymbol{x}) &= \int_0^\infty r^k f_{\boldsymbol{W}_{B_k}}(r\boldsymbol{x}_{B_k})\Pr(\boldsymbol{W}_{\overline{B}_k} \in [0, r\boldsymbol{x}_{\overline{B}_k}] \mid \boldsymbol{W}_{B_k} = r\boldsymbol{x}_{B_k})\mathrm{d}r \\
&= \int_0^\infty r^k \frac{\phi\left(\left(r\boldsymbol{x}^\circ_{B_k}\right)^{1/\nu}; \mathbf{0}_k, \Sigma_{B_kB_k}\right)}{\Phi^\circ(\boldsymbol{\infty}; \mathbf{0}, \Sigma)} \prod_{i=1}^k \left\{\frac{a_i}{\nu}(rx_i^\circ)^{1/\Delta - 1}\right\} \\
&\quad \times \Phi^\circ\left(\left(r\boldsymbol{x}^\circ_{\overline{B}_k}\right)^{1/\nu}; \Sigma_{\overline{B}_kB_k}\Sigma^{-1}_{B_kB_k}\left(r\boldsymbol{x}^\circ_{B_k}\right)^{1/\nu}, \Sigma_{\overline{B}_k\overline{B}_k} - \Sigma_{\overline{B}_kB_k}\Sigma^{-1}_{B_kB_k}\Sigma_{\overline{B}_kB_k}\right)\mathrm{d}r \\
&= \frac{\prod_{i\in B_k}\left\{x_i^{1/\nu - 1}a_i^{1/\nu}\right\}}{\Phi^\circ(\boldsymbol{\infty}; \mathbf{0}, \Sigma)}\nu^{-k}\int_0^\infty r^{k/\nu}\phi\left(\left(r\boldsymbol{x}^\circ_{B_k}\right)^{1/\nu}; \mathbf{0}_k, \Sigma_{B_kB_k}\right) \\
&\quad \times \Phi^\circ\left(\left(r\boldsymbol{x}^\circ_{\overline{B}_k}\right)^{1/\nu}; \Sigma_{\overline{B}_kB_k}\Sigma^{-1}_{B_kB_k}\left(r\boldsymbol{x}^\circ_{B_k}\right)^{1/\nu}, \Sigma_{\overline{B}_k\overline{B}_k} - \Sigma_{\overline{B}_kB_k}\Sigma^{-1}_{B_kB_k}\Sigma_{\overline{B}_kB_k}\right)\mathrm{d}r
\end{aligned}
$$

Using the notation $Q_\Sigma(\boldsymbol{x}) = \left\{\boldsymbol{x}^\top\Sigma^{-1}\boldsymbol{x}\right\}^{1/2}$ and applying the change of variable $t = r^{1/\nu}Q_{\Sigma_{B_kB_k}}(\boldsymbol{x}^\circ_{B_k})$, leads to

$$
\begin{aligned}
-V_{B_k}(\boldsymbol{x}) &= \frac{\prod_{i\in B_k}\left\{x_i^{1/\nu - 1}a_i^{1/\nu}\right\}}{\Phi^\circ(\boldsymbol{\infty}; \mathbf{0}, \Sigma)}\nu^{1-k}(2\pi)^{-k/2}|\Sigma_{B_kB_k}|^{-1/2}\int_0^\infty t^{k+\nu-1}Q_{\Sigma_{B_kB_k}}\left(\boldsymbol{x}^\circ_{B_k}\right)^{-k-\nu}e^{-t^2/2} \\
&\quad \times \Phi^\circ\left(\frac{\left(\boldsymbol{x}^\circ_{\overline{B}_k}\right)^{1/\nu}t}{Q_{\Sigma_{B_kB_k}}\left(\boldsymbol{x}^\circ_{B_k}\right)}; \frac{\Sigma_{\overline{B}_kB_k}\Sigma^{-1}_{B_kB_k}\left(\boldsymbol{x}^\circ_{B_k}\right)^{1/\nu}t}{Q_{\Sigma_{B_kB_k}}\left(\boldsymbol{x}^\circ_{B_k}\right)}, \Sigma_{\overline{B}_k\overline{B}_k} - \Sigma_{\overline{B}_kB_k}\Sigma^{-1}_{B_kB_k}\Sigma_{\overline{B}_kB_k}\right)\mathrm{d}t \\
&= \frac{\prod_{i\in B_k}\left\{x_i^{1/\nu - 1}a_i^{1/\nu}\right\}}{\Phi^\circ(\boldsymbol{\infty}; \mathbf{0}, \Sigma)}2^{(\nu-2)/2}\nu^{1-k}\pi^{-k/2}|\Sigma_{B_kB_k}|^{-1/2}Q_{\Sigma_{B_kB_k}}\left(\boldsymbol{x}^\circ_{B_k}\right)^{-k-\nu}\Gamma\left(\tfrac{k+\nu}{2}\right) \\
&\quad \times T^\circ_{\nu+k}\left(\frac{\left(\boldsymbol{x}^\circ_{\overline{B}_k}\right)^{1/\nu}}{Q_{\Sigma_{B_kB_k}}\left(\boldsymbol{x}^\circ_{B_k}\right)}; \frac{\Sigma_{\overline{B}_kB_k}\Sigma^{-1}_{B_kB_k}\left(\boldsymbol{x}^\circ_{B_k}\right)^{1/\nu}}{Q_{\Sigma_{B_kB_k}}\left(\boldsymbol{x}^\circ_{B_k}\right)}, \frac{\Sigma_{\overline{B}_k\overline{B}_k} - \Sigma_{\overline{B}_kB_k}\Sigma^{-1}_{B_kB_k}\Sigma_{\overline{B}_kB_k}}{k+\nu}\right),
\end{aligned}
$$

which concludes the proof. □

From Lemma 2, we can demonstrate that, for $k = 1, \ldots, D-1$ we have

$$\begin{aligned}
\lim_{\boldsymbol{x}_{\overline{B}_k} \downarrow \mathbf{0}_{D_k}} V_{B_k}(\boldsymbol{x}) &= \frac{\prod_{i \in B_k} \left\{a_i x_i^{1-\nu}\right\}^{1/\nu} 2^{(\nu-2)/2} \Gamma\left(\frac{k+\nu}{2}\right) \nu^{1-k}}{\Phi^{\circ}(\boldsymbol{\infty}, \mathbf{0}, \Sigma) \pi^{k/2} |\Sigma_{B_k B_k}|^{1/2} Q_{\Sigma_{B_k B_k}}\left(\boldsymbol{x}^{\circ}_{B_k}\right)^{k+\nu}} \\
&\quad \times \lim_{\boldsymbol{x}_{\overline{B}_k} \downarrow \mathbf{0}_{D_k}} T^{\circ}_{\nu+k}\left(\frac{\left(\boldsymbol{x}^{\circ}_{\overline{B}_k}\right)^{1/\nu}}{Q_{\Sigma_{B_k B_k}}\left(\boldsymbol{x}^{\circ}_{B_k}\right)}; \frac{\Sigma_{\overline{B}_k B_k} \Sigma^{-1}_{B_k B_k} \left(\boldsymbol{x}^{\circ}_{B_k}\right)^{1/\nu}}{Q_{\Sigma_{B_k B_k}}\left(\boldsymbol{x}^{\circ}_{B_k}\right)}, \frac{\Sigma_{\overline{B}_k \overline{B}_k} - \Sigma_{\overline{B}_k B_k} \Sigma^{-1}_{B_k B_k} \Sigma_{\overline{B}_k B_k}}{k+\nu} \right) \\
&= \frac{\prod_{i \in B_k} \left\{a_i x_i^{1-\nu}\right\}^{1/\nu} 2^{(\nu-2)/2} \Gamma\left(\frac{k+\nu}{2}\right) \nu^{1-k}}{\Phi^{\circ}(\boldsymbol{\infty}, \mathbf{0}, \Sigma) \pi^{k/2} |\Sigma_{B_k, B_k}|^{1/2} Q_{\Sigma_{B_k B_k}}\left(\boldsymbol{x}^{\circ}_{B_k}\right)^{k+\nu}} \\
&\quad \times T^{\circ}_{\nu+k}\left(\mathbf{0}_{D_k}; \frac{\Sigma_{\overline{B}_k B_k} \Sigma^{-1}_{B_k B_k} \left(\boldsymbol{x}^{\circ}_{B_k}\right)^{1/\nu}}{Q_{\Sigma_{B_k B_k}}\left(\boldsymbol{x}^{\circ}_{B_k}\right)}, \frac{\Sigma_{\overline{B}_k \overline{B}_k} - \Sigma_{\overline{B}_k B_k} \Sigma^{-1}_{B_k B_k} \Sigma_{\overline{B}_k B_k}}{k+\nu} \right) \\
&= \frac{\prod_{i \in B_k} \left\{a_i x_i^{1-\nu}\right\}^{1/\nu} 2^{(\nu-2)/2} \Gamma\left(\frac{k+\nu}{2}\right) \nu^{1-k}}{\Phi^{\circ}(\boldsymbol{\infty}, \mathbf{0}, \Sigma) \pi^{k/2} |\Sigma_{B_k, B_k}|^{1/2} Q_{\Sigma_{B_k B_k}}\left(\boldsymbol{x}^{\circ}_{B_k}\right)^{k+\nu}} \times 0 \\
&= 0,
\end{aligned}$$

which implies that all densities on $\partial\Omega$ take the value zero.

A.4 Exact simulation algorithm

The exact simulation algorithm for truncated extremal-t processes arises using Algorithm 2 in Dombry *et al.* (2016), which requires simulating a processes $\boldsymbol{W}/W(s_0)$ with distribution $P_{s_0}(A) = \mathrm{E}_{\boldsymbol{W}}[\mathbb{1}_{\{\boldsymbol{W}/W(s_0) \in A\}} W(s_0)]$ and where $\boldsymbol{W} = (W_0, W_1, \ldots, W_D) \equiv (W(s_0), W(s_1), \ldots, W(s_D))$ is defined as in Theorem 2 to the exception of being $D+1$ dimensional. The distribution P_{s_0} can be written as,

$$\begin{aligned}
P_{s_0}(A) &= \int_{\mathbb{R}_+^{D+1}} \mathbb{1}\left\{w_i/w_0 \in A_i, i = 1, \ldots, D\right\} w_0 \mathrm{d}F_{\boldsymbol{W}}(\boldsymbol{w}) \\
&= \int_{\mathbb{R}_+^{D+1}} \mathbb{1}\left\{w_i/w_0 \in A_i, i = 1, \ldots, D\right\} w_0 \frac{\phi\left((\boldsymbol{a}\boldsymbol{w})^{1/\nu}; \mathbf{0}_{D+1}, \Sigma\right)}{\Phi^{\circ}(\boldsymbol{\infty}; \mathbf{0}_{D+1}, \Sigma)} \prod_{i=0}^{D} (a_i w_i)^{1/\nu - 1} \frac{a_i}{\nu} \mathrm{d}\boldsymbol{w},
\end{aligned}$$

where $\boldsymbol{a} = (a_o, \ldots, a_D)$. Let $\boldsymbol{z} = (z_0, z_1, \ldots, z_D)$, with $z_0 = 1$ and $z_i = w_i/w_0, i = 1, \ldots, D$. The above equation can be written as

$$
\begin{aligned}
P_{s_0}(A) &= \int_{\mathbb{R}_+^{D+1}} \mathbb{1}\left\{z_i \in A_i, i = 1, \ldots, D\right\} w_0^{D+1} \frac{\phi\left((\boldsymbol{a}\boldsymbol{z})^{1/\nu}; \mathbf{0}_{D+1}, \Sigma\right)}{\Phi^{\circ}(\boldsymbol{\infty}; \mathbf{0}_{D+1}, \Sigma)} \prod_{i=0}^{D} (a_i w_0 z_i)^{1/\nu - 1} \frac{a_i}{\nu} \mathrm{d}w_0 d\boldsymbol{z}_{-1} \\
&= \int_{\mathbb{R}_+^{D+1}} \mathbb{1}\left\{z_i \in A_i, i = 1, \ldots, D\right\} w_0^{(D+1)/\nu} \frac{\phi\left((\boldsymbol{a}\boldsymbol{z})^{1/\nu}; \mathbf{0}_{D+1}, \Sigma\right)}{\Phi^{\circ}(\boldsymbol{\infty}; \mathbf{0}_{D+1}, \Sigma)} \prod_{i=0}^{D} (a_i z_i)^{1/\nu - 1} \frac{a_i}{\nu} \mathrm{d}w_0 d\boldsymbol{z}_{-1}
\end{aligned}
$$

Therefore, letting $\tilde{\boldsymbol{z}} = (\boldsymbol{a}\boldsymbol{z})^{1/\nu}$, the density of the distribution $P_{s_0}(\boldsymbol{z})$ is proportional to

$$
\begin{aligned}
f(\boldsymbol{z}) &\propto \prod_{i=1}^{D} \left\{z_i^{1/\nu - 1}\right\} \int_0^{\infty} w_0^{(D+1)/\nu} \exp\left\{-w_0^{2/\nu} \tilde{\boldsymbol{z}}^{\top} \Sigma^{-1} \tilde{\boldsymbol{z}}\right\} \mathrm{d}w_0 \\
&= \prod_{i=1}^{D} \left\{z_i^{1/\nu - 1}\right\} \int_0^{\infty} (2t)^{(D+1)/2} (\tilde{\boldsymbol{z}}^{\top} \Sigma^{-1} \tilde{\boldsymbol{z}})^{-(D+\nu+1)/2} e^{-t} \nu (2t)^{\nu/2 - 1} \mathrm{d}t \\
&\propto \prod_{i=1}^{D} \left\{z_i^{1/\nu - 1}\right\} \left(\tilde{\boldsymbol{z}}^{\top} \Sigma^{-1} \tilde{\boldsymbol{z}}\right)^{-(D+\nu+1)/2} \int_0^{\infty} t^{(D+\nu-1)/2} e^{-t} \mathrm{d}t \\
&\propto \prod_{i=1}^{D} \left\{z_i^{1/\nu - 1}\right\} \left(\tilde{\boldsymbol{z}}^{\top} \Sigma^{-1} \tilde{\boldsymbol{z}}\right)^{-(D+\nu+1)/2}
\end{aligned}
$$

Recall that B_D represents the set $\{1, \ldots, D\}$, given the block decomposition of the correlation matrix, $\Sigma = \begin{bmatrix} 1 & \Sigma_{0;B_D} \\ \Sigma_{0;B_D}^{\top} & \Sigma_{B_D;B_D} \end{bmatrix}$, the above equation can be written as

$$
f(\boldsymbol{z}) \propto \prod_{i=1}^{D} \left\{z_i^{1/\nu - 1}\right\} \left[\frac{\tilde{\boldsymbol{z}}_{B_D}/\tilde{z}_0 - \Sigma_{B_D;0}}{\Sigma_{B_D;B_D} - \Sigma_{B_D;0}\Sigma_{B_D;0}} \left(\tilde{\boldsymbol{z}}_{B_D}/\tilde{z}_0 - \Sigma_{B_D;0}\right) + 1\right]^{-(D+\nu+1)/2},
$$

which can be identified as the joint density function of $a_0 Z_i^{\nu}/a_i, i = 1, \ldots, D$, where $(Z_1, \ldots, Z_D)$ follows a multivariate truncated student-t distribution from below zero with scale parameter $(\Sigma_{B_D;B_D} - \Sigma_{B_D;0}\Sigma_{B_D;0}^{\top})/(\nu + 1)$, location parameter $\Sigma_{B_D;0}$ and $\nu + 1$ degrees of freedom. Algorithm 1 gives the pseudo-code of the exact simulation of the truncated extremal-t processes at finite location $s_1, \ldots, s_D$.

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Set Degrees of freedom $\nu > 0$ and some $D \times D$ correlation matrix Σ
Compute $a_{1:D}$ from (A.1).
Simulate $E_0 \sim \text{Exp}(1)$ and set $z = 1/E_0$.
Set $Y(s_1) = 1$ and simulate $Y(\boldsymbol{s})$ from a truncated-t distribution with location parameter $\Sigma_{-1;1}$, scale matrix $(\Sigma - \Sigma_{-1;1}\Sigma_{-1;1}^\top)/(\nu+1)$, and $\nu+1$ degrees of freedom at location $s_i \neq s_1$.
Set $W(s_j) = a_0 Y^\nu(s_j)/a_j$ for $j = 1, \ldots, D$.
Set $Z(s_j) = W(s_j)z$ for $j = 1, \ldots, D$.
For $i = 2, \ldots, D$
    Simulate $E_0 \sim \text{Exp}(1)$ and set $z = 1/E_0$.
    While $z > Z(\boldsymbol{s}_i)$
        Set $Y(s_j) = 1$ and simulate $Y(s_j)$, $j \neq i$ from a truncated-t distribution with location $\Sigma_{-j;j}$ and scale matrix $(\Sigma - \Sigma_{-j;j}\Sigma_{-j;j}^\top)/(\nu+1)$, and $\nu+1$ degrees of freedom.
        Let $W(s_j) = a_0 Y^\nu(s_j)/a_j$, $j = 1, \ldots, D$.
        If $W(s_j) * z < Z(\boldsymbol{s}_j), \forall j = 1, \ldots, i-1$
            Set $Z(s_j) = \max(Z(s_j), zW(s_j))$, $j = 1, \ldots, D$.
        Simulate $E_1 \sim \text{Exp}(1)$ and Set $E_0 = E_0 + E_1$ and $z = 1/E_0$.
Return $Z(s_j)$, $j = 1, \ldots, D$.

Algorithm 1: Exact simulation of truncated extremal-t processes at locations $\boldsymbol{s} = (s_1, \ldots, s_D)$

B Skew log-normal based max-stable processes

B.1 Proof of Theorem 3

By applying the first half of equation (8) in the main manuscript we have

$$\begin{aligned} V(\boldsymbol{x}) &= \sum_{k=1}^{D} \frac{1}{x_k} \int_{-\infty}^{\infty} \int_{-\infty}^{y_k + \boldsymbol{a}_{-k} - a_k \mathbf{1}_{D-1} + \log(\boldsymbol{x}_{-k}/x_k)} e^{y_k - a_k} \psi(\boldsymbol{y}; \mathbf{0}, \Sigma, \boldsymbol{\alpha}, 0) \mathrm{d}\boldsymbol{y}_{-k} \mathrm{d}y_k \\ &= \sum_{k=1}^{D} \frac{\Phi(\tau_k; 0, 1)}{2} \frac{\exp\{\tilde{\omega}_k^2/2 - a_k\}}{x_k} \int_{-\infty}^{\infty} \int_{-\infty}^{y_k + \boldsymbol{a}_{-k} - a_k \mathbf{1}_{D-1} + \log(\boldsymbol{x}_{-k}/x_k)} \psi(\boldsymbol{y}; \Sigma_{:,k}, \Sigma, \boldsymbol{\alpha}, \tau_k) \mathrm{d}\boldsymbol{y}_{-k} \mathrm{d}y_k \end{aligned} \tag{3}$$

where $\tau_k = \boldsymbol{\alpha}^\top \omega^{-1} \Sigma_{:,k} \left(1 + \boldsymbol{\alpha}^\top \overline{\Sigma} \boldsymbol{\alpha}\right)^{-1/2}$, $\overline{\Sigma} = \omega^{-1}\Sigma\omega^{-1}$, $\omega = \sqrt{\text{diag}(\Sigma)}$ and $\tilde{\boldsymbol{\omega}} = (\tilde{\omega}_1, \ldots, \tilde{\omega}_D) = \omega \mathbf{1}$. The above integrals represent $\Pr[\boldsymbol{X}_{-k} - X_k \leq \boldsymbol{a}_{-k} - a_k \mathbf{1}_{D-1} + \log(\boldsymbol{x}_{-k}/x_k)]$, with $\boldsymbol{X} \sim \text{ESN}(\Sigma_{:,k}, \Sigma, \boldsymbol{\alpha}, \tau_k)$. The random vector $\boldsymbol{X}_{-k} - X_k$ can be derived from $\boldsymbol{X}$ by applying the affine transformation $A_k \boldsymbol{X}$ where $A_k = (\boldsymbol{e}_1, \cdots, \boldsymbol{e}_{k-1}, -\mathbf{1}_{D-1}, \boldsymbol{e}_k, \ldots, \boldsymbol{e}_{D-1})$ and $\boldsymbol{e}_k, k = 1, \ldots, D-1$ are standard basis vectors of dimension $(D-1)$. From Azzalini (2013, Chapter 5) we have

$$\boldsymbol{X}_{-k} - X_k \sim \text{ESN}\left(\boldsymbol{\mu}_k, \Sigma_k, \widehat{\boldsymbol{\alpha}}_k, \tau_k\right)$$

where $\boldsymbol{\mu}_k = A_k\Sigma_{:,k}$, $\Sigma_k = A_k\Sigma A_k^\top$, $\widehat{\boldsymbol{\alpha}}_k = (1-\boldsymbol{\delta}^\top\omega A_k^\top\Sigma_k^{-1}A_k\omega\boldsymbol{\delta})^{-1/2}\widehat{\omega}_k\Sigma_k^{-1}A_k\omega\boldsymbol{\delta}$, $\boldsymbol{\delta} = \left(1+\boldsymbol{\alpha}^\top\overline{\Sigma}\boldsymbol{\alpha}\right)^{-1/2}\overline{\Sigma}\boldsymbol{\alpha}$, $\widehat{\omega}_k = \sqrt{\text{diag}(\Sigma_k)}$ and therefore, (3) becomes

$$V(\boldsymbol{x}) = \sum_{k=1}^{D} c\frac{\exp\{\tilde{\omega}_k^2/2-a_k\}}{x_k}\Psi\left(\boldsymbol{a}_{-k} - a_k\mathbf{1}_{D-1} + \log(\boldsymbol{x}_{-k}/x_k); \boldsymbol{\mu}_k, \Sigma_k, \widehat{\boldsymbol{\alpha}}_k, \tau_k\right). \tag{4}$$

The normalizing constants are derived by solving $a_k = \log\left(E\left[\exp\{Y_k\}\right]\right)$ for $k = 1,\ldots,D$. The moment generating function of $Y \sim \text{ESN}(\mathbf{0}, \Sigma, \boldsymbol{\alpha}, 0)$ is given by

$$M(\boldsymbol{t}) = 2\exp\left\{\tfrac{1}{2}\boldsymbol{t}^\top\Sigma\boldsymbol{t}\right\}\Phi\left(\boldsymbol{\delta}^\top\omega\boldsymbol{t}\right),$$

where $\boldsymbol{\delta}$ and ω are as defined above and considering $\boldsymbol{t} = \boldsymbol{e}_k$ yields

$$a_k = \log(2) + \tfrac{1}{2}\tilde{\omega}_k^2 + \log\left(\Phi\left(\tilde{\omega}_k\delta_k\right)\right).$$

Noting that $\tau_k = \tilde{\omega}_k\delta_k$ and

$$\boldsymbol{a}_{-k} - a_k\mathbf{1}_{D-1} + \log\left(\frac{\boldsymbol{x}_{-k}}{x_k}\right) = \log\left(\frac{\boldsymbol{x}_{-k}^{\circ\circ}}{x_k^{\circ\circ}}\right) + \frac{\tilde{\boldsymbol{\omega}}_{-k}^2 - \tilde{\omega}_k^2\mathbf{1}_{D-1}}{2},$$

where $\boldsymbol{x}^{\circ\circ} = \left(x_1^{\circ\circ},\ldots,x_D^{\circ\circ}\right)$, $x_k^{\circ\circ} = 2x_k\Phi\left(\tau_k;0,1\right)$, $k = 1,\ldots,D$, we can therefore re-write (4) as

$$\begin{aligned}
V(\boldsymbol{x}) &= \sum_{k=1}^{D}\frac{1}{x_k}\Psi\left(\log\left(\frac{\boldsymbol{x}_{-k}^{\circ\circ}}{x_k^{\circ\circ}}\right) + \frac{\tilde{\boldsymbol{\omega}}_{-k}^2 - \tilde{\omega}_k^2\mathbf{1}_{D-1}}{2}; \boldsymbol{\mu}_k, \Sigma_k, \widehat{\alpha}_k, \tau_k\right) \\
&= \sum_{k=1}^{D}\frac{1}{x_k}\frac{\Phi\left(\begin{bmatrix}\widehat{\omega}_k^{-1}\left(\log\left(\frac{\boldsymbol{x}_{-k}^{\circ\circ}}{x_k^{\circ\circ}}\right) + \frac{\tilde{\boldsymbol{\omega}}_{-k}^2 - \tilde{\omega}_k^2\mathbf{1}_{D-1}}{2} - \boldsymbol{\mu}_k\right) \\ \tau_k\end{bmatrix}; \begin{bmatrix}\mathbf{0}\\0\end{bmatrix}, \begin{bmatrix}\overline{\Sigma}_k & -\frac{\overline{\Sigma}_k\widehat{\boldsymbol{\alpha}}_k}{(1+\widehat{\boldsymbol{\alpha}}_k^\top\overline{\Sigma}_k\widehat{\boldsymbol{\alpha}}_k)^{1/2}} \\ -\frac{\widehat{\boldsymbol{\alpha}}_k^\top\overline{\Sigma}_k}{\left(1+\widehat{\boldsymbol{\alpha}}_k^\top\overline{\Sigma}_k\widehat{\boldsymbol{\alpha}}_k\right)^{1/2}} & 1\end{bmatrix}\right)}{\Phi(\tau_k;0,1)} \\
&= \sum_{k=1}^{D}\frac{1}{x_k}\frac{\Phi\left(\begin{bmatrix}\widehat{\omega}_k^{-1}\left(\log\left(\frac{\boldsymbol{x}_{-k}^{\circ\circ}}{x_k^{\circ\circ}}\right) + \frac{\tilde{\boldsymbol{\omega}}_{-k}^2 - \tilde{\omega}_k^2\mathbf{1}_{D-1}}{2} - \boldsymbol{\mu}_k\right) \\ \tau_k\end{bmatrix}; \begin{bmatrix}\mathbf{0}\\0\end{bmatrix}, \begin{bmatrix}\overline{\Sigma}_k & -\widehat{\omega}_k^{-1}A_k\omega\boldsymbol{\delta} \\ -\delta^\top\omega A_k^\top\widehat{\omega}_k^{-1} & 1\end{bmatrix}\right)}{\Phi(\tau_k;0,1)}.
\end{aligned}$$

which completes the proof of Theorem 3.

Notice that the expression above can be much simplified by re-parametrising the slant parameter $\boldsymbol{\alpha}$ by $\boldsymbol{\eta} = \omega^{-1}\boldsymbol{\alpha}$ which implies that $\omega\boldsymbol{\delta} = \Sigma\boldsymbol{\eta}\left(\sqrt{1+\boldsymbol{\eta}^\top\Sigma\boldsymbol{\eta}}\right)^{-1/2} \equiv \boldsymbol{\xi}$. As such, we have $\tau_k = \xi_k$

and $\widehat{\boldsymbol{\alpha}}_k = \left(1 - \boldsymbol{\xi}^\top A_k^\top \Sigma_k^{-1} A_k \boldsymbol{\xi}\right)^{-1/2} \widehat{\omega}_k \Sigma_k^{-1} A_k \boldsymbol{\xi}$, and the exponent function can be written as

$$V(\boldsymbol{x}) = \sum_{k=1}^{D} \frac{1}{x_k} \frac{\Phi\left(\begin{bmatrix} \widehat{\omega}_k^{-1}\left(\log\left(\frac{\boldsymbol{x}_{-k}^{\circ\circ}}{x_k^{\circ\circ}}\right) + \frac{\tilde{\boldsymbol{\omega}}_{-k}^2 - \tilde{\omega}_k^2 \mathbf{1}_{D-1}}{2} - \boldsymbol{\mu}_k\right) \\ \xi_k \end{bmatrix}; \begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix}, \begin{bmatrix} \overline{\Sigma}_k & -\widehat{\omega}_k^{-1} A_k \boldsymbol{\xi} \\ -\boldsymbol{\xi}^\top A_k^\top \widehat{\omega}_k^{-1} & 1 \end{bmatrix}\right)}{\Phi(\xi_k; 0, 1)}$$

$$= \sum_{k=1}^{D} \frac{1}{x_k} \frac{\Phi\left(\begin{bmatrix} \log\left(\frac{\boldsymbol{x}_{-k}^{\circ\circ}}{x_k^{\circ\circ}}\right) + \frac{\tilde{\boldsymbol{\omega}}_{-k}^2 - \tilde{\omega}_k^2 \mathbf{1}_{D-1}}{2} - \boldsymbol{\mu}_k \\ \xi_k \end{bmatrix}; \begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix}, \begin{bmatrix} \Sigma_k & -A_k \boldsymbol{\xi} \\ -\boldsymbol{\xi}^\top A_k^\top & 1 \end{bmatrix}\right)}{\Phi(\xi_k; 0, 1)}.$$

B.2 Proof of Proposition 2

The conditional probability in (12) from the manuscript is written as

$$\Pr(\boldsymbol{W}_{\overline{B}_k} = \mathbf{0}_{D-k} \mid \boldsymbol{W}_{B_k} = r\boldsymbol{x}_{B_k}) = \Pr(\boldsymbol{Y}_{\overline{B}_k} = -\boldsymbol{\infty}_{D-k} \mid \boldsymbol{Y}_{B_k} = \log(r\boldsymbol{x}_{B_k}) + a_{B_k}),$$

and since $\boldsymbol{Y}_{\overline{B}_k} \mid \boldsymbol{Y}_{B_k}$ is distributed according to the extended skew-Normal distribution, it directly implies that the above probability is zero and therefore the skewed Brown-Resnick process has no mass on $\partial\Omega$. It now remains to derive its densty on Ω°.

Take $\boldsymbol{w} = (w_1, \ldots, w_D) \in \mathbb{R}_+^D$ and let $w_k^{\circ\circ} = 2w_k \Phi(\tau_k; 0, 1)$, the density of $\boldsymbol{W}$ is given by

$$f_W(\boldsymbol{w}) = f_{\tilde{\boldsymbol{Y}}}\left(\frac{\tilde{\boldsymbol{w}}^2}{2} + \log\left(\boldsymbol{w}^{\circ\circ}\right)\right) \times \prod_{k=1}^{D} \frac{1}{w_k}$$

$$= \psi\left(\frac{\tilde{\boldsymbol{\omega}}^2}{2} + \log\left(\boldsymbol{w}^{\circ\circ}\right); \mathbf{0}, \Sigma, \boldsymbol{\alpha}, \mathbf{0}\right) \times \prod_{k=1}^{D} \frac{1}{w_k}. \tag{5}$$

In order to complete the proof we need the following lemma.

Lemma 3. *The intensity function for the skewed Brown-Resnick process is*

$$\kappa(\boldsymbol{x}) = \frac{2\Phi(\tilde{\tau}; 0, 1)|\Sigma|^{-1/2}\left(\mathbf{1}^\top \boldsymbol{q}\right)^{-1/2}}{(2\pi)^{(D-1)/2} \prod_{k=1}^{D} x_k} \exp\left\{-\frac{1}{2}\left[\log \boldsymbol{x}^{\circ\circ\top} \mathcal{A} \log \boldsymbol{x}^{\circ\circ} + \log \boldsymbol{x}^{\circ\circ\top}\left(\frac{2\boldsymbol{q}}{\mathbf{1}^\top \boldsymbol{q}} + \mathcal{A}\tilde{\boldsymbol{\omega}}^2\right) + \frac{\boldsymbol{q}^\top \tilde{\boldsymbol{\omega}}^2 - 1}{\mathbf{1}^\top \boldsymbol{q}} + \frac{1}{4}\tilde{\boldsymbol{\omega}}^{2,\top} \mathcal{A} \tilde{\boldsymbol{\omega}}^2\right]\right\},$$

where $\boldsymbol{q} = \Sigma^{-1}\mathbf{1}$, $\mathcal{A} = \Sigma^{-1} - \boldsymbol{q}\boldsymbol{q}^\top / \mathbf{1}^\top \boldsymbol{q}$ *and*

$$\tilde{\tau} = \left(1 + \frac{\left(\boldsymbol{\alpha}^\top \omega^{-1} \mathbf{1}\right)^2}{\mathbf{1}^\top q}\right)^{-1/2} \boldsymbol{\alpha}^\top \omega^{-1}\left[\left(\mathbb{I} - \frac{\mathbf{1} q^\top}{\mathbf{1}^\top q}\right)\left(\log \boldsymbol{x}^{\circ\circ} + \frac{\tilde{\boldsymbol{\omega}}^2}{2}\right) + \frac{\mathbf{1}}{\mathbf{1}^\top q}\right].$$

Proof. By plugging (5) into (9) from the main manuscript, we get

$$\begin{aligned}
\kappa(\boldsymbol{x}) &= \int_0^\infty r^D \psi\left(\frac{\tilde{\omega}^2}{2} + \log\left(\boldsymbol{x}^{\circ\circ}\right) + \log(r); \mathbf{0}, \Sigma, \boldsymbol{\alpha}, 0\right) \times \prod_{k=1}^{D} \frac{1}{r x_k} \mathrm{d}r \\
&= \int_0^\infty \psi\left(\frac{\tilde{\omega}^2}{2} + \log\left(\boldsymbol{x}^{\circ\circ}\right) + \log(r); \mathbf{0}, \Sigma, \boldsymbol{\alpha}, 0\right) \times \prod_{k=1}^{D} \frac{1}{x_k} \mathrm{d}r \\
&= \int_{\mathbb{R}} e^u \psi\left(\frac{\tilde{\omega}^2}{2} + \log\left(\boldsymbol{x}^{\circ\circ}\right) + u; \mathbf{0}, \Sigma, \boldsymbol{\alpha}, 0\right) \times \prod_{k=1}^{D} \frac{1}{x_k} \mathrm{d}u \\
&= \int_{\mathbb{R}} \frac{e^u 2}{\prod_{k=1}^{D} x_k} \phi\left(\frac{\tilde{\omega}^2}{2} + \log\left(\boldsymbol{x}^{\circ\circ}\right) + u; \mathbf{0}, \Sigma\right) \Phi\left(\boldsymbol{\alpha}^\top \omega^{-1}\left(\frac{\tilde{\omega}^2}{2} + \log\left(\boldsymbol{x}^{\circ\circ}\right) + u\right); 0, 1\right) \mathrm{d}u
\end{aligned}$$

Following Wadsworth and Tawn (2014), we have

$$\begin{aligned}
e^u \phi\left(\frac{\tilde{\omega}^2}{2} + \log\left(\boldsymbol{x}^{\circ\circ}\right) + u; \mathbf{0}, \Sigma\right) = \phi\left(u; \mu_u, \omega_u^2\right) & \frac{|\Sigma|^{-1/2}\left(\mathbf{1}^\top \boldsymbol{q}\right)^{-1/2}}{(2\pi)^{(D-1)/2}} \\
& \times \exp\left\{-\tfrac{1}{2}\left[\log \boldsymbol{x}^{\circ\circ\top} \mathcal{A} \log \boldsymbol{x}^{\circ\circ} + \log \boldsymbol{x}^{\circ\circ\top}\left(\frac{2\boldsymbol{q}}{\mathbf{1}^\top \boldsymbol{q}} + \mathcal{A}\tilde{\boldsymbol{\omega}}^2\right)\right]\right\} \\
& \times \exp\left\{-\tfrac{1}{2}\left(\frac{\boldsymbol{q}^\top \tilde{\boldsymbol{\omega}}^2 - 1}{\mathbf{1}^\top \boldsymbol{q}} + \tfrac{1}{4}\tilde{\boldsymbol{\omega}}^{2,\top} \mathcal{A} \tilde{\boldsymbol{\omega}}^2\right)\right\}
\end{aligned}$$

where $\boldsymbol{q} = \Sigma^{-1}\mathbf{1}$, $\mu_u = -\frac{\log \boldsymbol{x}^{\circ\circ}\boldsymbol{q} + \frac{1}{2}\left(\tilde{\boldsymbol{\omega}}^2\right)^\top \boldsymbol{q} - 1}{\mathbf{1}^\top \boldsymbol{q}}$, $\omega_u = \left(\mathbf{1}^\top \boldsymbol{q}\right)^{-1/2}$, and $\mathcal{A} = \Sigma^{-1} - \boldsymbol{q}\boldsymbol{q}^\top / \mathbf{1}^\top \boldsymbol{q}$. In addition, note that

$$\Phi\left(\boldsymbol{\alpha}^\top \omega^{-1}\left(\frac{\tilde{\omega}^2}{2} + \log\left(\boldsymbol{x}^{\circ\circ}\right) + u\right); 0, 1\right) = \Phi\left(\alpha_u\left(\frac{u - \mu_u}{\omega_u}\right) + \alpha_{0,u}; 0, 1\right) \frac{\Phi(\tilde{\tau}; 0, 1)}{\Phi(\tilde{\tau}; 0, 1)}$$

where $\alpha_u = \omega_u \boldsymbol{\alpha}^\top \omega^{-1}\mathbf{1}$, $\alpha_{0,u} = \boldsymbol{\alpha}^\top \omega^{-1}\left(I - \omega_u^2 \mathbf{1}\mathbf{1}^\top \Sigma^{-1}\right)\left(\log \boldsymbol{x}^{\circ\circ} + \tilde{\boldsymbol{\omega}}^2/2\right) - \omega_u^2 \boldsymbol{\alpha}^\top \omega^{-1}\mathbf{1}$ and

$$\tilde{\tau} = \alpha_{0,u}(1 + \alpha_u^2)^{-1/2} = \left(1 + \frac{(\boldsymbol{\alpha}^\top \omega^{-1}\mathbf{1})^2}{\mathbf{1}^\top \boldsymbol{q}}\right)^{-1/2} \boldsymbol{\alpha}^\top \omega^{-1}\left[\left(\mathbb{I} - \frac{\mathbf{1}\boldsymbol{q}^\top}{\mathbf{1}^\top \boldsymbol{q}}\right)\left(\log \boldsymbol{x}^{\circ\circ} + \frac{\tilde{\boldsymbol{\omega}}^2}{2}\right) + \frac{\mathbf{1}}{\mathbf{1}^\top \boldsymbol{q}}\right]. \quad (6)$$

Therefore, by matching the integrand with a univariate extended skew normal density function with

respect to u, we have,

$$
\begin{aligned}
\kappa(\boldsymbol{x}) = & \int_{\mathbb{R}} \phi\left(u; \mu_u, \omega_u^2\right) \Phi\left(\alpha_u\left(\tfrac{u-\mu_u}{\omega_u}\right) + \alpha_{0,u}; 0, 1\right) \tfrac{\Phi(\tilde{\tau};0,1)}{\Phi(\tilde{\tau};0,1)} \tfrac{2|\Sigma|^{-1/2}\left(\mathbf{1}^\top \boldsymbol{q}\right)^{-1/2}}{(2\pi)^{(D-1)/2}\prod_{k=1}^D x_k} \\
& \times \exp\left\{-\tfrac{1}{2}\left[\log \boldsymbol{x}^{\circ\circ\top} \mathcal{A} \log \boldsymbol{x}^{\circ\circ} + \log \boldsymbol{x}^{\circ\circ\top}\left(\tfrac{2\boldsymbol{q}}{\mathbf{1}^\top \boldsymbol{q}} + \mathcal{A}\tilde{\boldsymbol{\omega}}^2\right)\right]\right\} \\
& \times \exp\left\{-\tfrac{1}{2}\left(\tfrac{\boldsymbol{q}^\top \tilde{\boldsymbol{\omega}}^2 - 1}{\mathbf{1}^\top \boldsymbol{q}} + \tfrac{1}{4}\tilde{\boldsymbol{\omega}}^{2,\top} \mathcal{A} \tilde{\boldsymbol{\omega}}^2\right)\right\} \mathrm{d}u \\
= & \int_{\mathbb{R}} \phi\left(u; \mu_u, \omega_u^2\right) \Phi\left(\alpha_u\left(\tfrac{u-\mu_u}{\omega_u}\right) + \alpha_{0,u}; 0, 1\right) \tfrac{1}{\Phi(\tilde{\tau};0,1)} \mathrm{d}u \\
& \times \tfrac{2\Phi(\tilde{\tau};0,1)|\Sigma|^{-1/2}\left(\mathbf{1}^\top \boldsymbol{q}\right)^{-1/2}}{(2\pi)^{(D-1)/2}\prod_{k=1}^D x_k} \\
& \times \exp\left\{-\tfrac{1}{2}\left[\log \boldsymbol{x}^{\circ\circ\top} \mathcal{A} \log \boldsymbol{x}^{\circ\circ} + \log \boldsymbol{x}^{\circ\circ\top}\left(\tfrac{2\boldsymbol{q}}{\mathbf{1}^\top \boldsymbol{q}} + \mathcal{A}\tilde{\boldsymbol{\omega}}^2\right)\right]\right\} \\
& \times \exp\left\{-\tfrac{1}{2}\left(\tfrac{\boldsymbol{q}^\top \tilde{\boldsymbol{\omega}}^2 - 1}{\mathbf{1}^\top \boldsymbol{q}} + \tfrac{1}{4}\tilde{\boldsymbol{\omega}}^{2,\top} \mathcal{A} \tilde{\boldsymbol{\omega}}^2\right)\right\} \\
= & \tfrac{2\Phi(\tilde{\tau};0,1)|\Sigma|^{-1/2}\left(\mathbf{1}^\top \boldsymbol{q}\right)^{-1/2}}{(2\pi)^{(D-1)/2}\prod_{k=1}^D x_k} \exp\left\{-\tfrac{1}{2}\left[\log \boldsymbol{x}^{\circ\circ\top} \mathcal{A} \log \boldsymbol{x}^{\circ\circ} + \log \boldsymbol{x}^{\circ\circ\top}\left(\tfrac{2\boldsymbol{q}}{\mathbf{1}^\top \boldsymbol{q}} + \mathcal{A}\tilde{\boldsymbol{\omega}}^2\right) + \tfrac{\boldsymbol{q}^\top \tilde{\boldsymbol{\omega}}^2 - 1}{\mathbf{1}^\top \boldsymbol{q}} + \tfrac{1}{4}\tilde{\boldsymbol{\omega}}^{2,\top} \mathcal{A} \tilde{\boldsymbol{\omega}}^2\right]\right\},
\end{aligned}
$$

which gives the desired expression of the intensity function. □

By Lemma 3, the density on the interior of the simplex, Ω°, is therefore given by

$$
h(\boldsymbol{w}) = \tfrac{2\Phi(\tilde{\tau};0,1)|\Sigma|^{-1/2}\left(\mathbf{1}^\top \boldsymbol{q}\right)^{-1/2}}{(2\pi)^{(D-1)/2}\prod_{k=1}^D w_k} \exp\left\{-\tfrac{1}{2}\left[\log \boldsymbol{w}^{\circ\circ\top} \mathcal{A} \log \boldsymbol{w}^{\circ\circ} + \log \boldsymbol{w}^{\circ\circ\top}\left(\tfrac{2\boldsymbol{q}}{\mathbf{1}^\top \boldsymbol{q}} + \mathcal{A}\tilde{\boldsymbol{\omega}}^2\right) + \tfrac{\boldsymbol{q}^\top \tilde{\boldsymbol{\omega}}^2 - 1}{\mathbf{1}^\top \boldsymbol{q}} + \tfrac{1}{4}\tilde{\boldsymbol{\omega}}^{2,\top} \mathcal{A} \tilde{\boldsymbol{\omega}}^2\right]\right\},
$$

with $\tilde{\tau}$ as in (6) with $\boldsymbol{x}^{\circ\circ}$ replaced by $\boldsymbol{w}^{\circ\circ}$.

B.3 Partial derivatives of the exponent function

Lemma 4. *The partial derivatives of the exponent function for the skewed Brown-Resnick process are given as*

$$
\begin{aligned}
-V_{B_k}(\boldsymbol{x}) = & -\tfrac{V_{B_k}(\boldsymbol{x}_{B_k}, \boldsymbol{\infty}_{D-k})}{\Phi\left(\tilde{\alpha}_0\left(1+\tilde{\boldsymbol{\alpha}}^\top \overline{\tilde{\Sigma}} \tilde{\boldsymbol{\alpha}}\right)^{-1/2}; 0, 1\right)} \\
& \times \Phi\left(\begin{bmatrix} \tilde{\omega}_{(B_k)}^{-1}\left(\log \boldsymbol{x}_{B_k}^{\circ\circ} - \tilde{\boldsymbol{\mu}}\right) \\ \tilde{\alpha}_0(1 + \tilde{\boldsymbol{\alpha}}^\top \overline{\tilde{\Sigma}} \tilde{\boldsymbol{\alpha}})^{-1/2} \end{bmatrix}; \begin{bmatrix} \mathbf{0} \\ 0 \end{bmatrix} \begin{bmatrix} \overline{\tilde{\Sigma}} & -(1 + \tilde{\boldsymbol{\alpha}}^\top \overline{\tilde{\Sigma}} \tilde{\boldsymbol{\alpha}})^{-1/2} \overline{\tilde{\Sigma}} \tilde{\boldsymbol{\alpha}} \\ -(1 + \tilde{\boldsymbol{\alpha}}^\top \overline{\tilde{\Sigma}} \tilde{\boldsymbol{\alpha}})^{-1/2} \overline{\tilde{\Sigma}} \tilde{\boldsymbol{\alpha}} & 1 \end{bmatrix}\right),
\end{aligned}
$$

where $-V_{B_k}(\boldsymbol{x}_{B_k}, \boldsymbol{\infty}_{D-k})$ is the marginal intensity function, which equals the intensity function in Lemma 3 with parameters replaced by the marginal covariance matrix $\Sigma_{B_k B_k}$ and the marginal

slant parameter, $\boldsymbol{\alpha}_{(B_k)}$*, defined as,*

$$\boldsymbol{\alpha}_{(B_k)} = \left(1 + \boldsymbol{\alpha}_{\overline{B}_k}^{\top}\left(\overline{\Sigma}_{\overline{B}_k\overline{B}_k} - \overline{\Sigma}_{\overline{B}_k B_k}\overline{\Sigma}_{B_k B_k}^{-1}\overline{\Sigma}_{B_k\overline{B}_k}\right)\boldsymbol{\alpha}_{\overline{B}_k}\right)^{-1/2}\left(\boldsymbol{\alpha}_{B_k} + \overline{\Sigma}_{B_k B_k}^{-1}\overline{\Sigma}_{B_k\overline{B}_k}\boldsymbol{\alpha}_{\overline{B}_k}\right),$$

and $\tilde{\Sigma} = \left(\mathcal{A}_{\overline{B}_k\overline{B}_k}\right)^{-1}$, $\overline{\tilde{\Sigma}} = \omega_{(\overline{B}_k)}^{-1}\tilde{\Sigma}\omega_{(\overline{B}_k)}^{-1}$, $\omega_{(\overline{B}_k)} = \sqrt{\text{diag}(\tilde{\Sigma})}$, $\tilde{\boldsymbol{\alpha}} = \left(1 + \frac{(\alpha^{\top}\omega^{-1}\mathbf{1})^2}{\mathbf{1}^{\top}q}\right)^{-1/2}\omega_{(\overline{B}_k)}K_{01}\left(\mathbb{I} - \frac{\mathbf{1}q^{\top}}{\mathbf{1}^{\top}q}\right)\omega^{-1}\boldsymbol{\alpha}$,

$\tilde{\alpha}_0 = \boldsymbol{\beta}^{\top}K_{10}^{\top}(\log \boldsymbol{x}_{B_k}^{\circ\circ} + K_{10}\tilde{\boldsymbol{\omega}}^2/2) + \boldsymbol{\beta}^{\top}K_{01}^{\top}(\tilde{\boldsymbol{\mu}} + K_{01}\tilde{\boldsymbol{\omega}}^2/2) + \left(1 + \frac{(\boldsymbol{\alpha}^{\top}\boldsymbol{\omega}^{-1}\mathbf{1})^2}{\mathbf{1}^{\top}q}\right)^{-1/2}\frac{\boldsymbol{\alpha}^{\top}\boldsymbol{\omega}^{-1}\mathbf{1}}{\mathbf{1}^{\top}q}$, $K_{10} =$

$(\mathbb{I}_k, \mathbf{0}_{k,D-k})$, $K_{01} = (\mathbf{0}_{D-k,k}, \mathbb{I}_{D-k})$, $\boldsymbol{\beta} = \left(1 + \frac{(\boldsymbol{\alpha}^{\top}\omega^{-1}\mathbf{1})^2}{\mathbf{1}^{\top}q}\right)^{-1/2}\left(\mathbb{I} - \frac{q\mathbf{1}^{\top}}{\mathbf{1}^{\top}q}\right)\omega^{-1}\boldsymbol{\alpha}$, $\tilde{\boldsymbol{\mu}} = -\tilde{\Sigma}\left(\mathcal{A}_{\overline{B}_k B_k}\log \boldsymbol{x}_{B_k}^{\circ\circ} + K_{01}q/(\mathbf{1}^{\top}$

Proof. We can write

$$-V_{B_k}(\boldsymbol{x}) = -V_{B_k}(x_{B_k}, \boldsymbol{\infty}_{D-k})\int_{\mathbf{0}}^{\boldsymbol{x}_{\overline{B}_k}} \frac{V_{B_D}(\boldsymbol{x}_{B_k}, \boldsymbol{y})}{V_{B_k}(x_{B_k}, \boldsymbol{\infty}_{D-k})}\mathrm{d}\boldsymbol{y},$$

which involves two parts, the marginal intensity function $-V_{B_k}(x_{B_k}, \infty_{D-k})$ and the integral of the conditional intensity function over $\boldsymbol{x}_{\overline{B}_k}$. As the marginal distribution of the constructive spectral component $\boldsymbol{Y}_{B_k}$ in Theorem 3 from the main manuscript follows the distribution $\text{ESN}(\mathbf{0}, \Sigma_{B_k B_k}, \boldsymbol{\alpha}_{(B_k)}, \mathbf{0})$ (Azzalini, 2013, Chapter 5.1.4), where $\boldsymbol{\alpha}_{(B_k)}$ is defined above, the marginal intensity function $-V_{B_k}(x_{B_k}, \infty_{D-k})$ has the same expression as in Lemma 3 but with parameters, Σ and $\boldsymbol{\alpha}$, replaced by their marginal counterparts, $\Sigma_{B_k B_k}$ and $\boldsymbol{\alpha}_{(B_k)}$.

The conditional intensity function $V_{B_D}(\boldsymbol{x})/V_{B_k}(\boldsymbol{x}_{B_k}, \boldsymbol{\infty}_{D-k})$ is indeed a valid density function (Wadsworth and Tawn, 2013) with respect to $\boldsymbol{x}_{\overline{B}_k}$. Treating $\boldsymbol{x}_{B_k}$ as constant, the conditional intensity function can be expressed as

$$\begin{aligned}
&\frac{V_{B_D}(\boldsymbol{x})}{V_{B_k}(\boldsymbol{x}_{B_k}, \boldsymbol{\infty}_{D-k})} \\
&\propto \kappa(\boldsymbol{x}) \\
&= \frac{2\Phi(\tilde{\tau}; 0, 1)|\Sigma|^{-1/2}(\mathbf{1}^{\top}\boldsymbol{q})^{-1/2}}{(2\pi)^{(D-1)/2}\prod_{k=1}^{D}x_k}\exp\left\{-\frac{1}{2}\left[\log \boldsymbol{x}^{\circ\circ\top}\mathcal{A}\log \boldsymbol{x}^{\circ\circ} + \log \boldsymbol{x}^{\circ\circ\top}\left(\frac{2\boldsymbol{q}}{\mathbf{1}^{\top}\boldsymbol{q}} + \mathcal{A}\tilde{\boldsymbol{\omega}}^2\right) + \frac{\boldsymbol{q}^{\top}\tilde{\boldsymbol{\omega}}^2 - 1}{\mathbf{1}^{\top}\boldsymbol{q}} + \frac{1}{4}\tilde{\boldsymbol{\omega}}^{2,\top}\mathcal{A}\tilde{\boldsymbol{\omega}}^2\right]\right\}.
\end{aligned}$$

By comparing above equation with the multivariate extended skew normal density function, we can show that the conditional intensity function with respect to $\log \boldsymbol{x}_{\overline{B}_k}^{\circ\circ}$ is a density function for a multivariate extended skew normal random vector, $\log \boldsymbol{X}_{\overline{B}_k}^{\circ\circ}$, with scale matrix $\tilde{\Sigma} = \left(\mathcal{A}_{\overline{B}_k\overline{B}_k}\right)^{-1}$, location

parameter $\tilde{\boldsymbol{\mu}} = -\tilde{\Sigma}\left(\mathcal{A}_{\overline{B}_k B_k}\log(\boldsymbol{x}^{\circ\circ}_{B_k}) + K_{01}q/(\mathbf{1}^\top q) + K_{01}\mathcal{A}\tilde{\boldsymbol{\omega}}^2/2\right)$, where $K_{01} = (\mathbf{0}_{D-k,k}, \mathbb{I}_{D-k})$, slant parameter $\tilde{\boldsymbol{\alpha}} = \left(1 + \frac{(\alpha^\top\omega^{-1}\mathbf{1})^2}{\mathbf{1}^\top q}\right)^{-1/2}\omega_{(\overline{B}_k)}K_{01}\left(\mathbb{I} - \frac{\mathbf{1}q^\top}{\mathbf{1}^\top q}\right)\omega^{-1}\boldsymbol{\alpha}$, where $\omega_{(\overline{B}_k)} = \sqrt{\mathrm{diag}(\tilde{\Sigma})}$, and parameter

$$\tilde{\alpha}_0 = \boldsymbol{\beta}^\top K_{10}^\top(\log \boldsymbol{x}^{\circ\circ}_{B_k} + K_{10}\tilde{\boldsymbol{\omega}}^2/2) + \boldsymbol{\beta}^\top K_{01}^\top(\tilde{\boldsymbol{\mu}} + K_{01}\tilde{\boldsymbol{\omega}}^2/2) + \left(1 + \frac{(\alpha^\top\omega^{-1}\mathbf{1})^2}{\mathbf{1}^\top q}\right)^{-1/2}\frac{\boldsymbol{\alpha}^\top\boldsymbol{\omega}^{-1}\mathbf{1}}{\mathbf{1}^\top q},$$

where $K_{10} = (\mathbb{I}_k, \mathbf{0}_{k,D-k})$ and $\boldsymbol{\beta} = \left(1 + \frac{(\alpha^\top\omega^{-1}\mathbf{1})^2}{\mathbf{1}^\top q}\right)^{-1/2}\left(\mathbb{I} - \frac{q\mathbf{1}^\top}{\mathbf{1}^\top q}\right)\omega^{-1}\boldsymbol{\alpha}$. Therefore, we have

$$\begin{aligned}\frac{V_{B_D}(\boldsymbol{x})}{V_{B_k}(\boldsymbol{x}_{B_k}, \boldsymbol{\infty}_{D-k})} &= \frac{|\tilde{\Sigma}|^{-1/2}(2\pi)^{-(D-k)/2}}{\Phi\left(\tilde{\alpha}_0\left(1+\tilde{\boldsymbol{\alpha}}^\top\omega^{-1}_{(\overline{B}_k)}\tilde{\Sigma}\omega^{-1}_{(\overline{B}_k)}\tilde{\boldsymbol{\alpha}}\right)^{-1/2};0,1\right)\prod_{i\in\overline{B}_k}x_i^{\circ\circ}}\Phi\left(\tilde{\alpha}_0 + \tilde{\boldsymbol{\alpha}}^\top\omega^{-1}_{(\overline{B}_k)}\left(\log \boldsymbol{x}^{\circ\circ}_{\overline{B}_k} - \tilde{\boldsymbol{\mu}}\right);0,1\right)\\ &\quad\times\exp\left\{-\tfrac{1}{2}(\log \boldsymbol{x}^{\circ\circ}_{\overline{B}_k} - \tilde{\boldsymbol{\mu}})^\top\mathcal{A}_{\overline{B}_k\overline{B}_k}(\log \boldsymbol{x}^{\circ\circ}_{\overline{B}_k} - \tilde{\boldsymbol{\mu}})\right\}.\end{aligned}$$

Then, by using the cumulative distribution function of $\log \boldsymbol{X}^{\circ\circ}_{\overline{B}_k}$, the integral can be computed as

$$\begin{aligned}\int_{\mathbf{0}}^{\boldsymbol{x}_{\overline{B}_k}}\frac{V_{B_D}(\boldsymbol{x}_{B_k},\boldsymbol{y})}{V_{B_k}(x_{B_k},\boldsymbol{\infty}_{D-k})}\mathrm{d}\boldsymbol{y} &= \Pr[\boldsymbol{X}_{\overline{B}_k} \le \boldsymbol{x}_{\overline{B}_k}]\\ &= \Pr[\log \boldsymbol{X}_{\overline{B}_k} \le \log \boldsymbol{x}_{\overline{B}_k}]\\ &= \Pr[\log \boldsymbol{X}^{\circ\circ}_{\overline{B}_k} \le \log \boldsymbol{x}^{\circ\circ}_{\overline{B}_k}]\\ &= \frac{\Phi\left(\begin{bmatrix}\tilde{\omega}^{-1}_{(\overline{B}_k)}(\log \boldsymbol{x}^{\circ\circ}_{\overline{B}_k} - \tilde{\boldsymbol{\mu}})\\ \tilde{\alpha}_0(1+\tilde{\boldsymbol{\alpha}}^\top\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}})^{-1/2}\end{bmatrix};\begin{bmatrix}\mathbf{0}\\0\end{bmatrix},\begin{bmatrix}\overline{\tilde{\Sigma}} & -(1+\tilde{\boldsymbol{\alpha}}^\top\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}})^{-1/2}\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}}\\ -(1+\tilde{\boldsymbol{\alpha}}^\top\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}})^{-1/2}\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}} & 1\end{bmatrix}\right)}{\Phi(\tilde{\alpha}_0(1+\tilde{\boldsymbol{\alpha}}^\top\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}})^{-1/2})}\end{aligned}$$

where $\overline{\tilde{\Sigma}} = \omega^{-1}_{(\overline{B}_k)}\tilde{\Sigma}\omega^{-1}_{(\overline{B}_k)}$. Plugging the above results, the partial derivative of the exponent function are given by

$$\begin{aligned}-V_{B_k}(\boldsymbol{x}) &= \frac{-V_{B_k}(\boldsymbol{x}_{B_k},\boldsymbol{\infty}_{D-k})}{\Phi(\tilde{\alpha}_0(1+\tilde{\boldsymbol{\alpha}}^\top\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}})^{-1/2})}\\ &\quad\times\Phi\left(\begin{bmatrix}\tilde{\omega}^{-1}_{(\overline{B}_k)}(\log \boldsymbol{x}^{\circ\circ}_{\overline{B}_k} - \tilde{\boldsymbol{\mu}})\\ \tilde{\alpha}_0(1+\tilde{\boldsymbol{\alpha}}^\top\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}})^{-1/2}\end{bmatrix};\begin{bmatrix}\mathbf{0}\\0\end{bmatrix},\begin{bmatrix}\overline{\tilde{\Sigma}} & -(1+\tilde{\boldsymbol{\alpha}}^\top\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}})^{-1/2}\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}}\\ -\left(1+\tilde{\boldsymbol{\alpha}}^\top\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}}\right)^{-1/2}\overline{\tilde{\Sigma}}\tilde{\boldsymbol{\alpha}} & 1\end{bmatrix}\right)\end{aligned}$$

which concludes the proof. □

From Lemma 4, one can easily show the limit of the partial derivative $-V_{B_k}(\boldsymbol{x})$ for any index set $B_k \neq \emptyset$ and $B_k \subsetneq \{1,\ldots,D\}$ as $\boldsymbol{x}_{\overline{B}_k} \to \mathbf{0}$ is zero.

B.4 Exact simulation algorithms

Let $\boldsymbol{W} = (W_0, W_1, \ldots, W_D) \equiv (W(s_0), W(s_1), \ldots, W(s_D))$ be defined as in Theorem 3 to the exception of being $D+1$ dimensional. The distribution P_{s_0} defines the joint distribution of $W_i/W_0, i = 1, \ldots, D$ under the transformed probability measure $\widehat{\Pr} = W(s_0)\mathrm{dPr}$. The moment generating function of $Y_i \equiv Y(s_i), i = 0, \ldots, D$ evaluated at $\boldsymbol{t} = (t_0, t_1, \ldots, t_D)$ can be calculated under the transformed probability measure $\widehat{\Pr}$ as

$$\begin{aligned}\mathrm{E}\left[\exp\left\{Y_0 - a_0 + \sum_{i=0}^{D} t_i Y_i\right\}\right] &= 2\exp\left\{\tfrac{1}{2}(1+t_0, \boldsymbol{t}_{B_D})^\top \Sigma (1+t_0, \boldsymbol{t}_{B_D}) - a_0\right\} \Phi(\delta^\top \omega (1+t_0, \boldsymbol{t}_{B_D})) \\ &= 2\exp\{\tfrac{1}{2}(\Sigma_{0;0} + \boldsymbol{t}^\top \Sigma \boldsymbol{t}) + \Sigma_{0;\cdot}\boldsymbol{t} - a_0\} \Phi\left(\delta_0\omega_{0;0} + \delta^\top \omega \boldsymbol{t}\right)\end{aligned}$$

where the matrices ω and Σ can be decomposed as $\Sigma = \begin{bmatrix} \Sigma_{00} & \Sigma_{B_D;0} \\ \Sigma_{B_D;0}^\top & \Sigma_{B_D;B_D} \end{bmatrix}$. Plugging in the expression for a_0 given in Theorem 3 leads to

$$\mathrm{E}\left[\exp\left\{Y_0 - a_0 + \sum_{i=0}^{D} t_i Y_i\right\}\right] = \exp\{\tfrac{1}{2}\boldsymbol{t}^\top \Sigma \boldsymbol{t} + \Sigma_{\cdot;0}\boldsymbol{t}^\top\} \frac{\Phi(\delta_0\omega_{00} + \delta^\top \omega \boldsymbol{t})}{\Phi(\delta_0\omega_{00})},$$

which corresponds to the moment generating function of the extended skew-normal random vector $(Y^*(s_0), \ldots, Y^*(s_D))$ with location parameter $\Sigma_{\cdot;0}$, scale parameter Σ, slant parameter $\boldsymbol{\alpha}$, and the extended parameter $\tau_0 = \delta_0\omega_{00}$. The distribution of P_{s_0} therefore becomes

$$\exp\{Y^*(s_i) - Y^*(s_0) - a_i + a_0\}, i = 0, \ldots, D.$$

Details about random generation from the multivariate extended skew-normal can be found in (Azzalini, 2013, Chapter 5.3.3). The pseudo-code of the exact simulation algorithms is summarised in Algorithm 2.

.

Set the slant parameter $\boldsymbol{\alpha} \in \mathbb{R}^D$ and the $D \times D$ scale matrix Σ
Compute $\boldsymbol{a}$ and δ as defined in Theorem 3.
Simulate $E_0 \sim \text{Exp}(1)$ and set $z = 1/E_0$.
Simulate $Y^*(\boldsymbol{s})$ from an extended skew normal with location parameter $\Sigma_{:1}$,
scale matrix Σ, slant parameter $\boldsymbol{\alpha}$ and the extended parameter $\tau_1 = \delta_1\omega_{11}$.
Set $W(\boldsymbol{s}) = \exp(Y(\boldsymbol{s}) - Y(s_1) + a_1 - \boldsymbol{a})$
Set $Z(s_j) = W(s_j) * z$ for $j = 1, \ldots, D$.
For $i = 2, \ldots, D$,
    Simulate $E_0 \sim \text{Exp}(1)$ and set $z = 1/E_0$.
    While $z > Z(\boldsymbol{s}_i)$
        Simulate $Y(\boldsymbol{s})$ from an extended skew normal with location parameter $\Sigma_{:i}$,
        scale matrix Σ, slant parameter $\boldsymbol{\alpha}$ and extended parameter $\tau_i = \delta_i\omega_{ii}$
        Set $W(\boldsymbol{s}) = \exp(\boldsymbol{Y}(s) - Y(s_i) + a_i - \boldsymbol{a})$
        If $W(s_j) * z < Z(\boldsymbol{s}_j), \forall j = 1, \ldots, i-1$
            Set $Z(s_j) = \max(Z(s_j), W(s_j) * z), j = 1, \ldots, D$.
        Simulate $E_1 \sim \text{Exp}(1)$, Set $E_0 = E_0 + E_1$ and $z = 1/E_0$.
Return $Z(s_j), j = 1, \ldots, D$.

Algorithm 2: Exact simulation from the skewed Brown-Resnick process at location $s_i, i = 1, \ldots, D$

C Proof of Theorem 1

For any $k \in \{1, \ldots, D-1\}$ we have

$$\begin{aligned}\lim_{\boldsymbol{x}_{\overline{B}_k} \downarrow \mathbf{0}_{D-k}} -V_{B_k}(\boldsymbol{x}) &= \int_0^\infty r^k f_{\boldsymbol{W}_{B_k}}(r\boldsymbol{x}_{B_k}) \lim_{\boldsymbol{x}_{\overline{B}_k} \downarrow \mathbf{0}_{D-k}} \Pr(\boldsymbol{W}_{\overline{B}_k} \in [\mathbf{0}, r\boldsymbol{x}_{\overline{B}_k}] \mid \boldsymbol{W}_{B_k} = r\boldsymbol{x}_{B_k}) \mathrm{d}r \\ &= \int_0^\infty r^k f_{\boldsymbol{W}_{B_k}}(r\boldsymbol{x}_{B_k}) \Pr(\boldsymbol{W}_{\overline{B}_k} = \mathbf{0}_{D-k} \mid \boldsymbol{W}_{B_k} = r\boldsymbol{x}_{B_k}) \mathrm{d}r,\end{aligned}$$

which implies that if $\Pr(\boldsymbol{W}_{\overline{B}_k} = \mathbf{0}_{D-k} \mid \boldsymbol{W}_{B_k} = r\boldsymbol{x}_{B_k}) = 0$ then $\lim_{\boldsymbol{x}_{\overline{B}_k} \downarrow \mathbf{0}_{D-k}} -V_{1:k}(\boldsymbol{x}) = 0$ and consequently by (**??**), $h_{B_k}\left(\boldsymbol{x}_{B_k}/\|\boldsymbol{x}_{B_k}\|\right) = 0$. Moreover, if the angular density on $\partial\Omega$ is zero everywhere, then we must have

$$\lim_{x_{\overline{B}_k} \downarrow \mathbf{0}_{D-k}} -V_{B_k}(\boldsymbol{x}) = \int_0^\infty r^k f_{\boldsymbol{W}_{B_k}}(r\boldsymbol{x}_{B_k}) \Pr(W_{\overline{B}_k} = \mathbf{0}_{D-k} \mid \boldsymbol{W}_{Bk} = r\boldsymbol{x}_{B_k}) \mathrm{d}r = 0.$$

As the integrand is non-negative and the integral is zero, we conclude that the integrand must be zero almost everywhere. Therefore, we have $\Pr(W_{\overline{B}_k} = \mathbf{0}_{D-k} \mid W_{B_k} = r\boldsymbol{x}_{B_k}) = 0$ for any $r > 0$ and

$\boldsymbol{x}_{B_k} > \mathbf{0}_k$ almost everywhere because r^k is positive and W is a non-negative continuous random process, which implies the condition in (**??**) must hold.

D Proof of Theorem 4

The distribution of the r-Pareto process with risk functional r is

$$\kappa_D(\cdot \cap \mathcal{A}_r)/\kappa_D(\mathcal{A}_r).$$

Suppose we have $\tilde{\boldsymbol{Y}} \in \Omega$ follow the angular distribution $\mathcal{H}/D$ and $\tilde{R}$ follows a unit Pareto distribution, i.e., $\Pr[\tilde{R} > x] = 1/x, x > 1$, we have

$$\Pr[\tilde{R}\tilde{\boldsymbol{Y}} \in \tilde{A}] = \tfrac{1}{D}\int_\Omega \int_1^\infty x^{-2}\mathbf{1}_{\{x\boldsymbol{w}\in\tilde{A}\}}\mathrm{d}x\mathcal{H}(\mathrm{d}\boldsymbol{w}).$$

If we have $\tilde{A} \subset \mathcal{A}_{r_0}$, then $x\boldsymbol{w} \in \tilde{A}$ leads to $r_0(x\boldsymbol{w}) = x\|w\|_1 = x > 1$. Therefore, we have

$$\Pr[\tilde{R}\tilde{\boldsymbol{Y}} \in \tilde{A}] = \tfrac{1}{D}\int_\Omega \int_1^\infty x^{-2}\mathbf{1}_{\{x\boldsymbol{w}\in\tilde{A}\}}\mathrm{d}x\mathcal{H}(\mathrm{d}\boldsymbol{w}) = \tfrac{1}{D}\int_\Omega \int_0^\infty x^{-2}\mathbf{1}_{\{x\boldsymbol{w}\in\tilde{A}\}}\mathrm{d}x\mathcal{H}(\mathrm{d}\boldsymbol{w}) = \kappa_D(\tilde{A})/D.$$

As we have $\Pr(\tilde{R}\tilde{\boldsymbol{Y}} \in \mathcal{A}_{r_0}) = \kappa_D(\mathcal{A}_{r_0})/D = 1/D$, it immediately follows that $\tilde{R}\tilde{\boldsymbol{Y}}$ is distributed as the r-Pareto process with risk functional r_0. Let $\tilde{r}(\cdot) = c_0 r(\cdot)$, then $\tilde{r}$ will be also a risk functional that is convex and homogeneous of order 1. Thus, we have $\tilde{r}(\boldsymbol{e}_i) = c_0/c_i \leq 1$. The complement set of $\mathcal{A}_{\tilde{r}}$ in $\mathbb{R}_+^D$, $\mathcal{A}_{\tilde{r}}^C = \{\boldsymbol{x} \in \mathbb{R}_+^D : \tilde{r}(\boldsymbol{x}) \leq 1\}$, is a convex set and contains points $\boldsymbol{e}_i, i = 1, \ldots, D$ on its boundary as well as the origin $\mathbf{0}$ since any homogenous function of order 1 maps the origin to the origin. For r_0, we know that $\mathcal{A}_{r_0}^C$ is a convex set and contains $\mathbf{0}$ and $\boldsymbol{e}_i, i = 1, \ldots D$ on its boundary, and $\mathcal{A}_{r_0}^C$ is the smallest convex set that contains points $\boldsymbol{e}_i, i = 1, \ldots, D$ and $\mathbf{0}$. Therefore, we have $\mathcal{A}_{r_0}^C \subset \mathcal{A}_{\tilde{r}}^C$, which implies that $\mathcal{A}_{\tilde{r}} \subset \mathcal{A}_{r_0}$. If we have $\tilde{A} \subset \mathcal{A}_{\tilde{r}}$, then we also have $\tilde{A} \subset \mathcal{A}_{r_0}$. Moreover, we can get

$$\Pr[\tilde{\boldsymbol{Z}}^{(r_0)} \in \tilde{A} | \tilde{\boldsymbol{Z}}^{(r_0)} \in \mathcal{A}_{\tilde{r}}] = \frac{1/D\kappa_D(\tilde{A}\cap\mathcal{A}_{\tilde{r}})}{1/D\kappa_D(\mathcal{A}_{\tilde{r}})} = \frac{\kappa_D(\tilde{A}\cap\mathcal{A}_{\tilde{r}})}{\kappa_D(\mathcal{A}_{\tilde{r}})} \tag{7}$$

Since κ_D is homogeneous of order -1 and $\mathcal{A}_{\tilde{r}} = 1/c_0\mathcal{A}_r$, it follows that

$$\begin{aligned}
&\Pr[\tilde{\boldsymbol{Z}}^{(r_0)} \in \tilde{A} | \tilde{\boldsymbol{Z}}^{(r_0)} \in \mathcal{A}_{\tilde{r}}] \\
&= \Pr[c_0\tilde{\boldsymbol{Z}}^{(r_0)} \in c_0\tilde{A} | c_0\tilde{\boldsymbol{Z}}^{(r_0)} \in \mathcal{A}_r] \\
&= \frac{\kappa_D(\tilde{A}\cap\mathcal{A}_r/c_0)}{\kappa_D(\mathcal{A}_r/c_0)} = \frac{c_0\kappa_D(c_0\tilde{A}\cap\mathcal{A}_r)}{c_0\kappa_D(\mathcal{A}_r)} = \frac{\kappa_D(c_0\tilde{A}\cap\mathcal{A}_r)}{\kappa_D(\mathcal{A}_r)}.
\end{aligned}$$

where $c_0\tilde{A} \subset \mathcal{A}_r$ since $\tilde{A} \subset \mathcal{A}_{\tilde{r}}$. Therefore, we have $\tilde{\boldsymbol{Z}}^{(r)} \overset{\mathrm{d}}{=} c_0\tilde{\boldsymbol{Z}}^{(r_0)} \mid r\left(c_0\tilde{\boldsymbol{Z}}^{(r_0)}\right) > 1$.

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